## 1. Name of the Student

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## 2. Title

**Timer-Based LED Flasher PCB** (Assignment Question 6)

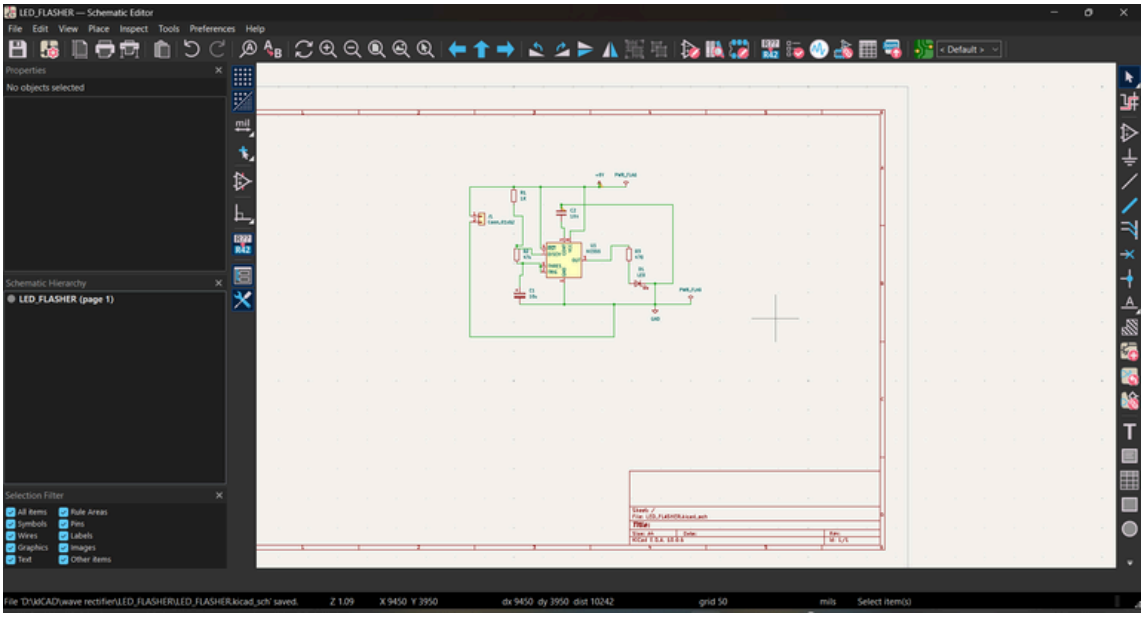
## Design overview

A signage company needs a simple circuit that periodically flashes an LED to attract attention on information displays and warning signs. The design uses an NE555 timer in astable mode driving an LED through a current-limiting resistor, powered from a 9 V supply via a 2-pin connector. The board was designed in KiCad (schematic, footprint assignment, layout and routing).

## Frequency calculation

$f = 1.44 / ((R1 + 2 \cdot R2) \cdot C1) = 1.44 / ((1\ k\Omega + 2 \times 47\ k\Omega) \times 10\ \mu F) = 1.44 / 0.95 \approx$ **1.5 Hz**
High time $t_H = 0.693\ (R1 + R2)\ C1 \approx 0.33$ s; low time $t_L = 0.693\ R2\ C1 \approx 0.33$ s; duty cycle ≈ 51%.
LED current: $I = (V_{OUT} - V_F) / R3\ (9 - 1.5 - 2.0) / 470 \approx 12$ mA (safe for a standard 5 mm LED).

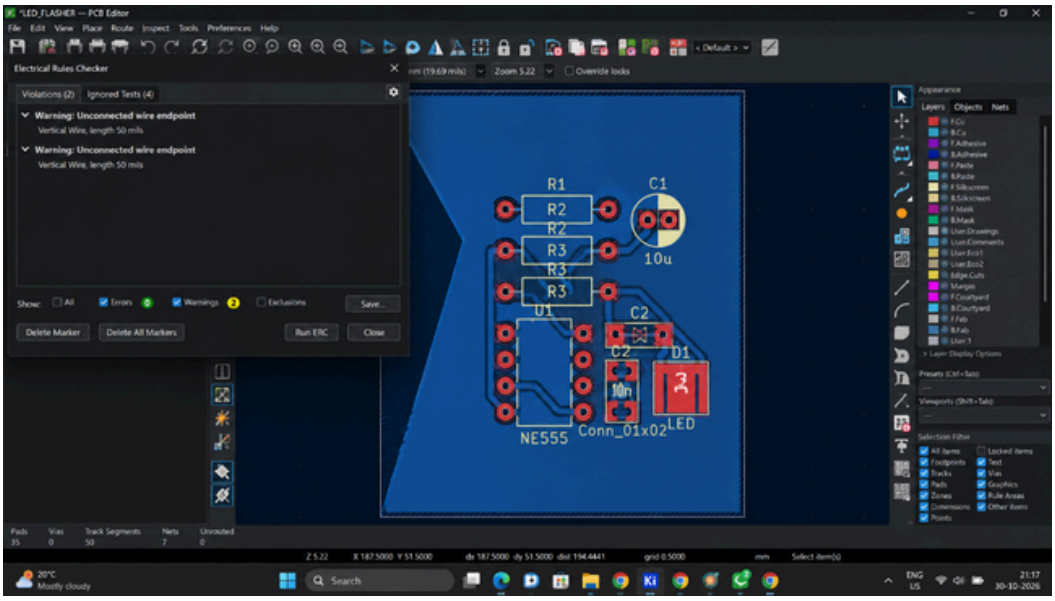
# 3. Schematic




## Netlist / connections

| Net | Connections |
|---|---|
| +9V | J1-1, U1 pin 8 (VCC), U1 pin 4 (RESET), R1 |
| GND | J1-2, U1 pin 1, C1(-), C2, D1 cathode |
| N_DISCH | R1, R2, U1 pin 7 (DISCH) |
| N_TIMING | R2, U1 pin 6 (THRES), U1 pin 2 (TRIG), C1(+) |
| N_CTRL | U1 pin 5 (CTRL), C2 |
| N_OUT | U1 pin 3 (OUT), R3 |
| N_LED | R3, D1 anode |

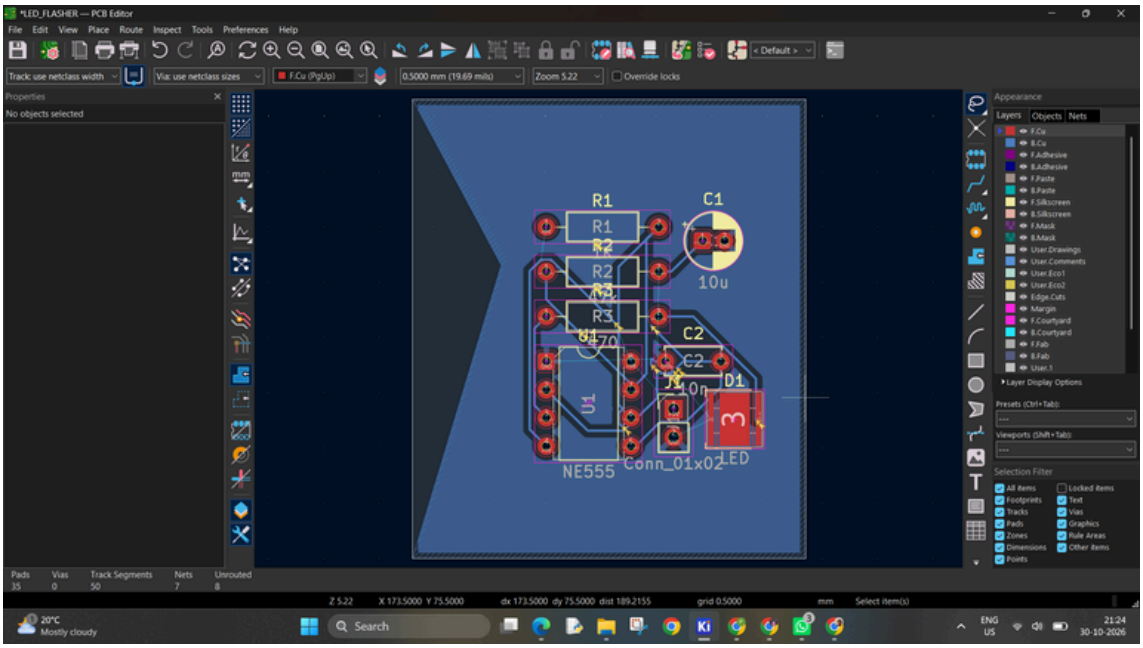
# 4. Screenshot of Footprint After Routing Is Completed



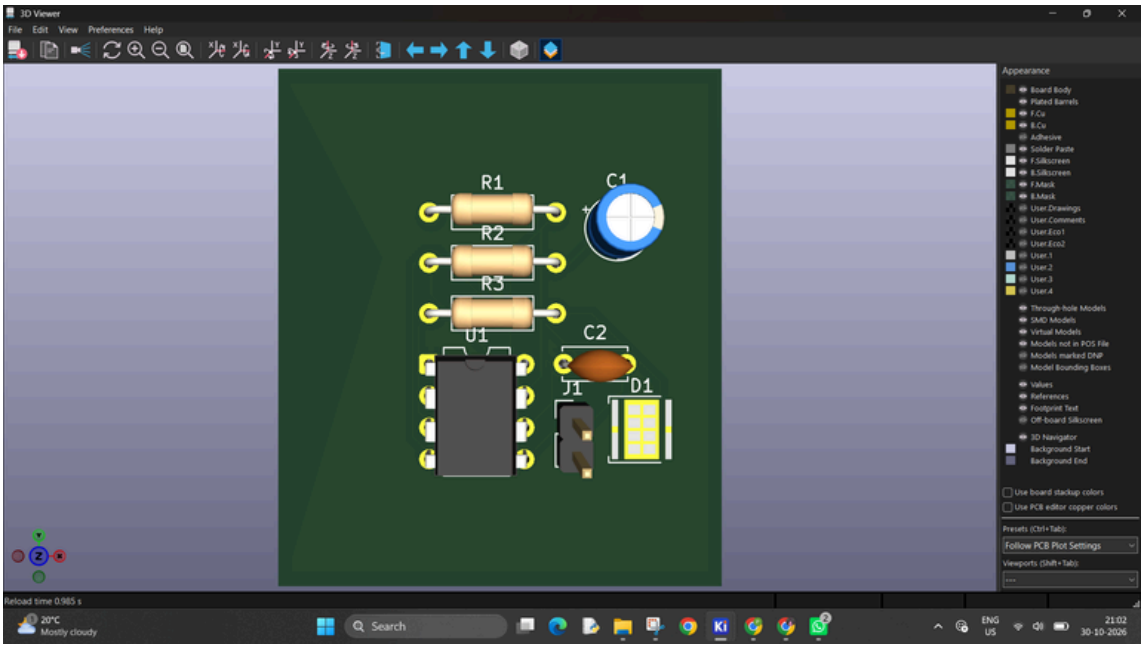

## Footprint assignment

| Ref | Footprint |
|---|---|
| U1 | Package_DIP:DIP-8_W7.62mm |
| R1, R2, R3 | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal |
| C1 | Capacitor_THT:CP_Radial_D5.0mm_P2.00mm |
| C2 | Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm |
| D1 | LED_THT:LED_D5.0mm |
| J1 | Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical |

# 5. 2D PCB View


*LED_FLASHER — PCB Editor
File Edit View Place Route Inspect Tools Preferences Help
R1
C1
10u
R2
R3
C2
D1
NE555
Conn_01x02
LED

# 6. 3D PCB View


3D Viewer
File Edit View Preferences Help
R1
R2
R3
C1
C2
U1
J1
D1
Appearance
Search

# 7. Bill of Materials (BOM)

| Ref | Qty | Value | Footprint | Description |
|---|---|---|---|---|
| U1 | 1 | NE555 | DIP-8_W7.62mm | 555 timer IC |
| R1 | 1 | 1 k | R_Axial_DIN0207 P10.16 | Timing resistor |
| R2 | 1 | 47 k | R_Axial_DIN0207 P10.16 | Timing resistor |
| R3 | 1 | 470 | R_Axial_DIN0207 P10.16 | LED current limiting |
| C1 | 1 | 10 uF | CP_Radial_D5.0mm_P2.00mm | Timing capacitor (electrolytic) |
| C2 | 1 | 10 nF | C_Disc_D5.0mm_P5.00mm | Control-pin bypass |
| D1 | 1 | LED red | LED_D5.0mm | Flasher LED |
| J1 | 1 | Conn_01x02 | PinHeader_1x02_P2.54mm | 9 V power input |

Total components: 9 (8 line items).